

The raster model, and what follows from it

A value for each cell
of a regular grid; location
calculated from the grid

reading a file

terra::rast() **terra::inMemory()**
the header is a few numbers whatever the size of the file, so the values stay on disk; the source line is a filename until an operation puts the result in memory

the size of a cell

terra::res() **terra::cellSize()**
res() is in the units of the CRS, so it is an angle in a geographic one; cellSize() calculates area on the ground after accounting for the CRS

so a failing cell is
emptied, not removed

filter()
empties the failing cells rather than removing them, so ncell() does not move and a count of cells stops being a count of values

combining measurements

c() **select()** **rename()** **mutate()**
c() stacks rasters that share a grid, matched by position; the verbs then address layers the way their dplyr forms address columns

when two axes are
not enough

read_stars()
a cube with a named dimension for every axis, useful when time, depth or spectral band is central to the analysis

writing a file

terra::writeRaster()
datatype sets the bits per cell and truncates the values that do not fit; compression changes file size without changing cell values